ChatGPT

Executive Summary

This document defines a **shared U.S. grazing-land suitability framework** for cow-calf, sheep, and goat systems. Its purpose is **early-stage land screening** rather than exact stocking or financial analysis. The architecture is layered:

- **Shared Land Factors** – a unified set of physical and ecological factors (soil, terrain, vegetation, water, climate) describing the parcel, independent of livestock.
- **Species-Specific Rules** – separate rule sets for cattle, sheep and goats that interpret the same land factors differently (e.g. high shrub cover may be unfavorable for cattle but beneficial for goats).
- **Regional/Ecological Context** – parameters or modifiers (like climate, ecoregion or NRCS ecological site) that calibrate the base rules without hard-coding state-by-state differences.

This ensures **scalability**: core livestock biology is universal, while climate and soil inputs capture location differences. For the MVP, Texas is the first **validation sandbox** (due to its large cattle, sheep and goat industries【3†L214-L223】【8†L2-L9】), but no rule is Texas-only. The same factors and rules apply nationwide, subject to local context.

We recommend a final set of **10 shared core factors** (IDs F01–F10) that all three profiles use, with species-specific interpretations. Each factor is supported by scientific sources (USDA/NRCS/extension) and mapped to U.S. datasets. A deterministic matching engine will score each parcel–profile pair for these factors, then compare results across species and flag unknowns or constraints.

Key outputs include a decision label (`ADVANCE/REVIEW/HOLD/REDIRECT/REJECT`), suitability summary, signals, known/unknown statuses, and next-step recommendations. Below we detail the factors, data sources, rules and validation approach.

Shared Core Factors (IDs F01–F10)

We propose these **10 core factors** for the MVP. (Operational/other factors like fencing or legal access are handled separately in due diligence.)

ID	Factor	Role
F01	**Terrain & Slope**	Topography influences grazing distribution
F02	**Herbaceous (Grass/Forb) Cover**	Abundance of grass/forb forage
F03	**Shrub / Woody Browse Cover**	Abundance of shrubs/browse
F04	**Herbaceous Production (Forage Signal)**	Estimated forage biomass signal
F05	**Soil / Site Water Capacity**	Soil and hydrology underpin vegetation
F06	**Livestock Water Availability**	Presence of usable water points
F07	**Livestock Water Distribution**	Distance/geography of water relative to forage

ID	Factor	Role
F08	**Precipitation (Climatic Input)**	Long-term rainfall regime
F09	**Drought Exposure / Variability**	Historical moisture deficits and variability
F10	**Flood / Wetness Exposure**	Frequency/extent of flooding or ponding

Each factor is defined below with scientific rationale, measurement, species-specific relationships, data sources, and rule examples.

Factor Details

F01 – Terrain & Slope

Definition: Topography of the parcel (slope, elevation, ruggedness) that affects animal movement and usable area.

Measurable Variables: Mean and max slope; elevation bands; ruggedness indices; proportion of area in steep classes; rock or outcrop presence.

Species Relationships:

- **Cattle:** Strongly negative beyond gentle slopes. Cattle prefer flats and lower slopes; they "have trouble walking and grazing on steep slopes" 【1†L39-L41】. Typically they avoid slopes >20–30%.
- **Sheep:** Moderately negative only on extreme terrain. Sheep are more agile and often use steeper areas (up to ~45% slopes in some systems【3†L219-L223】).
- **Goats:** Least negatively affected; goats are very agile climbers and will use rough, steep terrain readily【3†L219-L223】. However, very steep ground still reduces accessible area and may harbor predators or unsafe footing.

Importance: High for all systems as it directly limits usable pasture. Critical for distinguishing cattle (flat-preferring) from goats (terrain-tolerant).

Data Sources:

- **USGS/DEM:** 3DEP/NED elevation data (10m-30m, national coverage) or SRTM (30m) for slope/elevation.
- **Mireye (remote imagery):** Possible terrain mosaic or contour layers.
- **NRCS SSURGO:** "Slope" attribute (percentage in each soil map unit).

Remote-Measurability: High (topography is well-mapped). Mireye-derived slope maps can augment DEM data.

Limitations:

- **No Single Threshold:** Slope tolerances can vary by ecological site and stocking density. Avoid global cutoffs (e.g. ">30% = no cattle") without context.
- **Vegetation Interaction:** Steepness often correlates with vegetation type (e.g. savanna vs. forest). Interpretation should consider combined factors.

Verification Needs: On-site checks if slope-index alone flags issues (e.g. steep slopes with poor footing or erosion might need in-person review).

Rule Template (Deterministic Examples):

```
IF avg_slope > X% THEN
   Cattle_Suitability = Low;
ELSE
   Cattle_Suitability = High;
```

```
IF %area >30% slope  > 20% THEN
   Sheep_Caution = true; Goat_Suitability = High;
```

(Specific thresholds X should be calibrated per region/ecosystem.)

F02 – Herbaceous (Grass/Forb) Cover

Definition: Fractional cover of herbaceous vegetation (grasses and forbs) on the parcel.

Measurable Variables: Percent cover of annual grasses, perennial grasses, annual forbs, perennial forbs. Total herbaceous cover.

Species Relationships:

- **Cattle:** Strong positive. Cattle diets are ~70% grass on average【3†L198-L202】. High grass cover indicates abundant feed.
- **Sheep:** Positive. Sheep use a mix of grass and forbs (~50% grass【3†L198-L202】) but also forbs; grass cover still largely correlates with total forage.
- **Goats:** Moderate positive. Goats rely more on browse, but still need some herbaceous grazing. Low grass cover reduces total available forage.

Importance: High. Indicates baseline forage availability and complements production metrics.

Data Sources:

- **USDA-ARS Rangeland Analysis Platform (RAP):** Annual layers for "annual grasses/forbs cover" and "perennial grasses/forbs cover" across U.S. rangelands【3†L198-L202】.
- **Mireye imagery:** Vegetation segmentation might estimate herbaceous fraction.
- **NDVI/EVI:** as proxy (less interpretable by class).

Remote-Measurability: High for cover fraction (modern sensors and RAP provide this data). Less certain on palatability.

Limitations:

- **Cover ≠ Forage Quality:** A high cover of unpalatable weeds (or dead biomass) may not help livestock.

- **Seasonality:** RAP' s annual measures can be timed to peak biomass, but grazing capacity varies seasonally.
- **Understory (forbs) vs. grasses:** Combined in cover data, may need species rules to parse.

Verification Needs: Field check for forage quality/composition (e.g. invasive grasses, poisonous weeds).

Rule Template:

```
IF herbaceous_cover low THEN
   Cattle_Suitability = Low; Sheep_Suitability = Low;
```

```
IF herbaceous_cover high THEN
   Goat_Suitability = Neutral (since goats eat browse too);
```

F03 – Shrub / Woody Browse Cover

Definition: Fractional cover of shrubs, brush and other woody plants.

Measurable Variables: Percent shrub cover; brush canopy height/density; presence of browse species.

Species Relationships:

- **Cattle:** Usually negative. High shrub cover often displaces grasses or blocks grazing. Cattle prefer open pastures.
- **Sheep:** Variable. Moderate shrub cover can provide supplementary browse, but too much dense brush reduces grass.
- **Goats:** Often positive. Goats are browsers; more available brush means more food. However, see **limitations** below.

This factor is critical to distinguish systems. **Example:** "Same land fact, different meaning: high brush cover penalizes cattle but can boost goat potential【3†L198-L202】【3†L221-L224】."

Importance: Medium–High. Key differentiator factor (goats vs. cattle/sheep).

Data Sources:

- **USDA-ARS RAP:** "Shrub cover" and "tree cover" layers (annual estimates).
- **Mireye:** High-res segmentation might identify brush clusters or woody vegetation lines.

Remote-Measurability: High for total structural cover, but not for species or nutritional quality.

Limitations:

- **Browse Quality Unknown:** Remote data cannot tell if shrubs are palatable or toxic (e.g. some goats eat oak/acacia, others avoid it).
- **Hidden Browse:** Taller trees vs. low browse both count as "shrub cover" in RAP.
- **Zero Shrub ≠ Good Goats:** A lack of shrubs means goats have no browse, but that may just mean they feed more on herbs.

Verification Needs: On-site survey of browse species (palatable vs. toxic), fence safety through brush, etc.

Rule Template:

```
IF shrub_cover high AND plant_composition_allows THEN
   Goat_Suitability = High;
ELSE IF shrub_cover high THEN
   Note: Goat_Suitability may be *Conditionally Good* (verify species).
```

```
IF shrub_cover > Y% THEN
   Cattle_Suitability = Lower; Goat_Suitability = Higher;
```

F04 – Herbaceous Production (Forage-Production Signal)

Definition: Estimated annual herbaceous biomass (forage production) – a **productivity signal**, not literal carrying capacity.

Measurable Variables: Above-ground biomass (e.g. lbs/acre) from remote-sensing models; separate annual vs. perennial production.

Species Relationships:

- **Cattle:** Very High importance. Adequate total forage is required for viable cow-calf production.
- **Sheep:** High. Total forage correlates with carrying capacity.
- **Goats:** Medium–High. Goats supplement grass with browse, but overall plant productivity still limits stocking.

Importance: Very high. Direct proxy for how much vegetation the land can grow.

Data Sources:

- **USDA-ARS RAP:** Provides modeled "Herbaceous Production" (annual forage biomass) layers across U.S. grazing lands. (catalog.data.gov).
- **MODIS/NDVI time series:** for anomaly/drought detection (less precise absolute).

Remote-Measurability: Medium–High. RAP gives modeled estimates, but they rely on calibration.

Limitations:

- **Not Carrying Capacity:** Does not account for seasonality, grazing utilization rate, or ecological site use.
- **Quality vs. Quantity:** High biomass with low quality (e.g. mature grass) may not support the same stocking.
- **Ecological Site Dependence:** A sandy site vs. clay site will respond differently to rainfall; absolute production values require context.

Verification Needs: Ground-truthing with field measurements or expert knowledge, particularly if RAP indicates unexpectedly low production.

Rule Template:

```
IF annual_production_signal low THEN
   All_Species_Suitability = Low (insufficient forage).
```

```
IF production_signal high THEN
   Cattle_Suitability = High (assuming grass quality); else Sheep/Goat similarly boosted.
```

F05 – Soil & Site Water-Holding Characteristics

Definition: Underlying soil and hydrological properties determining moisture availability (e.g., texture, depth, drainage, available water capacity).

Measurable Variables: Soil texture classes (sand/silt/clay), drainage class (well, moderately, poorly drained), available water-holding capacity (AWC), effective root depth, flood frequency (soil attribute).

Species Relationships: All three species depend on how soils and topography affect forage potential and stress:

- Poorly drained or shallow soils may limit forage on all systems, as roots and plants can’t develop well.
- Conversely, soils with high water-holding (clay loams vs. coarse sand) can buffer drought stress.

Importance: High (indirectly). Soils set the ecological potential of the land. NRCS uses Ecological Site Descriptions combining soil+climate to define forage capacity【4†L132-L141】.

Data Sources:

- **NRCS SSURGO Database:** Soil survey maps with attributes (drainage class, AWC, pH, depth, flood frequency).
- **NRCS Ecological Site Descriptions (ESD):** Derived attributes for broad site potential (may guide rule calibration).

Remote-Measurability: High for mapped attributes (SSURGO coverage is national, though resolution varies by area). Lower for actual on-ground variation.

Limitations:

- **Site-Specific Complexity:** Soils vary at fine scale; parcel may span several soil mapunits.
- **Vegetation Baseline:** Soil properties must be interpreted in context (e.g. shallow soil + high rainfall might still produce ample forage).
- **No Direct Scores:** Hard to say “good” or “bad” soil for goats vs. cattle; better used in combined calculation or as a qualifier.

Verification Needs: May require soil sampling or NRCS field checks, especially if mapping shows mixed or marginal soils.

Rule Template:

```
IF soil_drainage = “poorly drained” THEN
   All_Species_Caution = true (verify root health and wetness).
```

```
IF available_water_capacity < threshold AND annual_precip < X THEN
   All_Species_Suitability = Reduced.
```

F06 – Livestock Water Availability

Definition: Presence of accessible water sources (ponds, streams, wells, troughs) that livestock can reliably use.

Measurable Variables: Number and type of mapped water features (perennial streams, ponds, springs, reservoirs, wells), distance to nearest water, presence of water infrastructure (pipes, troughs), groundwater info.

Species Relationships:

- **All species:** Essential. Cattle must drink daily; sheep/goats slightly less but still need frequent water. No grazing can succeed without assured water.
- **Cattle:** High water need (e.g., over 20 gallons/day per cow) and limited range (typically <1.5 mi from water)【3†L219-L223】.
- **Sheep/Goats:** More efficient. Goats/sheep will travel farther to water and require less per head.

Importance: Very High for all. A “physical constraint” layer – absence of water can override forage suitability.

Data Sources:

- **USGS NHD (National Hydrography Dataset):** Mapped streams, lakes and ponds.
- **OpenStreetMap / local data:** May include wells or ponds.
- **Mireye:** High-res imagery analysis could detect ponds, tanks, troughs (if visible).
- **Public Water Index / Aerial Imagery:** For surface water detection.

Remote-Measurability: Medium. Large water bodies are obvious, but small stock tanks or seasonal creeks may be missed. Reliable flows cannot be assured remotely.

Limitations:

- **Not Reliability:** Knowing there is a pond doesn’t ensure it has year-round water, good quality, or legal water rights.
- **False Negatives:** Lack of mapped water is not proof of no water (could be an uncharted well or intermittently wet area).
- **No Threshold:** “At least 1 water source per X acres” is context-dependent.

Verification Needs: On-site or record review for well depth, pump capacity, water rights and seasonal flow.

Rule Template:

```
IF no_mapped_water AND no_recorded_well THEN
   Suitability = HOLD (verification needed).
```

```
IF water_points >= 1 per 640 acres THEN
   Water_Sufficiency = OK; else Flag_Low_Water_Coverage.
```

F07 – Livestock Water Distribution

Definition: Spatial relationship of grazing areas to water (e.g. distance from forage to nearest water points).

Measurable Variables: Distance-to-water metrics (max distance, average distance, % of area beyond X miles from water); waterpoint density.

Species Relationships:

- **Cattle:** Negative impact if forage is far from water. Cattle seldom graze >1.5 miles from water 【3†L219-L223】, so large unwatered patches reduce usable acreage.
- **Sheep:** Less constrained; often will travel 2–3 miles.
- **Goats:** Even more independent; can cover rugged terrain to reach water.

Importance: High. Water distribution can effectively shrink grazing area.

Data Sources:

- **Derived:** Use location of water features (from F06 sources) and compute distances over the parcel. GIS or Mireye tools.
- **RAP / WorldClim:** Not directly; must compute with mapped data.

Remote-Measurability: Medium. Distance is computable, but true "usable water" points are uncertain (see F06).

Limitations:

- **As-the-crow-flies vs. actual travel:** Terrain may make short distances functionally longer.
- **Infrastructure:** If pipelines or troughs exist (not visible remotely), animal movement may differ.
- **Thresholds Vary:** "Good" distribution depends on stocking density and species mobility.

Verification Needs: Validate with field survey or known ranch practices. Could incorporate stockmanship (e.g. rotation) into final eval.

Rule Template:

```
IF max_distance_to_water > 2 mi AND species = Cattle THEN
   Suitability = Lower (flag distant grazing zone).
```

```
IF >50% area > 1 mi from water THEN
   All_Species -> “Potential grazing gap” (requires water development).
```

F08 – Precipitation (Climatic Input)

Definition: Precipitation regime (rainfall amounts, seasonality, and norms) of the location.

Measurable Variables: Long-term annual precipitation (30-year normals), seasonal patterns (e.g. wet season vs dry season), recent annual totals, and precipitation anomalies.

Species Relationships: Indirect for all. Precipitation sets moisture input for forage growth. Low-rainfall areas support less biomass; high rainfall areas allow more production but also more pasture mix (trees vs grass).

Importance: High as contextual driver. For example, Texas A&M Extension notes “Rainfall is a major limiting factor for livestock production from Texas rangelands” 【8†L2-L9】. However, precipitation should **inform forage expectations**, not be a direct score by itself.

Data Sources:

- **NOAA NCEI / PRISM:** High-resolution climate normals and records per county or grid.
- **USGS Climate Data:** Surface water supply index.
- **WorldClim/Daymet:** for free global interpolation.

Remote-Measurability: High (gridded climate is available).

Limitations:

- **Not a Hard Suitability Score:** Rain itself is not “good” or “bad” without context. We must use it to modulate vegetation expectations (higher rain → more forage, but also more risk of flooding or swamp).
- **Inter-annual Variability:** Single dry year may not doom a good site, but repeated droughts do (see F09).
- **Regional Differences:** 20″/yr is plenty in Montana, but minimal in Florida.

Verification Needs: Compare to local ecological site data or county extents.

Rule Template:

```
IF annual_precip < region_minimum THEN
   All_Species_Suitability = Lower (flag arid site).
```

```
IF precipitation_anomaly negative (drought) THEN
   Reduce all forage signals accordingly.
```

F09 – Drought Exposure & Variability

Definition: Historical frequency and severity of drought (relative to local climate), indicating risk and resilience.

Measurable Variables: Drought index (e.g. % years below X percentile rainfall, Palmer Drought Severity Index records, Vegetation production anomalies). Measures of interannual variability.

Species Relationships: All require stable forage. Frequent drought means reduced reliability. It does not inherently favor one species but may influence stocking strategy.

Importance: High as a risk factor. For instance, multi-year droughts can cause persistent reductions in forage quantity 【6 † L23-L32】. All grazing systems must plan for drought; high drought exposure lowers suitability confidence.

Data Sources:

- **NOAA/NCEI Drought Monitor or PDSI:** Historical maps and indices.
- **USDA Veg Drought (NASS):** Data on forage health.
- **RAP Time Series:** Yearly production can show variability.

Remote-Measurability: High (data available), but interpretation requires caution (e.g. a single bad year vs. general trend).

Limitations:

- **Temporal Scope:** Short-term drought doesn’t always translate to failed ranch; multi-year trends matter more.
- **Local Mitigation:** Infrastructure (e.g. wells, supplemental feeding) can offset drought.
- **Not a Suitability Ban:** Drought exposure should guide risk labeling, not automatic rejection.

Verification Needs: Check for historical herd records, drought relief measures, or adaptive management capability.

Rule Template:

```
IF 5-year drought_count > threshold THEN
   All_Species_Risk = High (recommend *REVIEW*).
```

```
IF region has frequent drought (e.g., <70% normal precipitation 3+ years/decade) THEN
   Mark Unknown/Needs Verification for long-term viability.
```

F10 – Flooding / Ponding Exposure

Definition: Tendency of parts of the parcel to be flooded or waterlogged (seasonally or permanently), reducing grazing availability.

Measurable Variables: % area in 100-year floodplain, soil flooding frequency class (NRCS), FEMA flood maps, historical inundation patterns.

Species Relationships: Negative for all (livestock avoid deep water), but moderate flooding may recharge pastures.

- **Cattle:** Averse to extremely wet ground (risk of hoof issues, inaccessible forage).
- **Sheep/Goats:** Sheep avoid flooded areas (similar hoof concerns); goats can wade brush but still avoid deep floods.

Importance: Medium (contextual). Flood zones reduce usable area but can create fertile grasslands.

Data Sources:

- **FEMA Flood Maps:** 100-year floodplain shapefiles (cover major rivers and coasts).
- **NRCS SSURGO:** Frequency/Duration of flooding for soil units.
- **USGS/GNIS:** Waterbody extents in wet seasons.

Remote-Measurability: Medium. FEMA maps exist but are coarse. Lidar DEM can help identify depressions.

Limitations:

- **Seasonal Variability:** An area flooded in spring might be dry summer grazing land.
- **Not Always Negative:** Some floodplain grazing is used (e.g. silage after dew).
- **Edge Cases:** Swamps, bogs or irrigation canals not always mapped.

Verification Needs: Local flood history, interviews, check for required flood-zone mitigation (e.g. grazers may seasonally avoid).

Rule Template:

```
IF %area in floodplain > 30% THEN
   Cattle_Suitability = Lower (risk of losing grazing area);
   Sheep/Goat_Suitability = Slightly Lower.
```

```
IF soil_map_flood_frequency = frequent AND species = Cattle THEN
   Suitability_Caution = true.
```

Factor–Data Mapping Table

The table below maps each core factor to suggested U.S. data sources and fields. “Mireye” refers to high-resolution remote imagery/indices that the RangeMatch system can analyze (when available). “Coverage” notes if the data is national or statewide.

Factor	Mireye/Imagery (example fields)	USDA-ARS RAP / Veg. Databases	NRCS/Soil Data	USGS/ Hydrography/ FEMA/NOAA
F01 Terrain & Slope	Digital Elevation Model (DEM) derived slope/ terrain; Lidar contours	N/A	SSURGO: “slope” (%) in each soil unit; hydrologic group	USGS 3DEP DEM (1″ or 1/3″), SRTM; 30m resolution (national)
F02 Herbaceous Cover	Vegetation segmentation from imagery; NDVI/EVI indices	RAP “Annual Grasses/Forbs Cover” and “Perennial Grasses/ Forbs Cover”; USGS NVC	N/A	N/A
F03 Shrub Cover	Shrub canopy detection (object-based classification on imagery)	RAP “Shrub Cover”, “Tree Cover” layers	N/A	N/A
F04 Herbaceous Production	NDVI time-series; grass height estimates (Mireye)	RAP “Herbaceous Production” (lbs/ acre)	N/A	NOAA Precip & Veg indices (for anomaly)
F05 Soil Properties	Soil proxies (e.g. vegetation reflectance/ texture)	N/A	SSURGO attributes: drainage class, available water capacity (AWC), flooding frequency, effective depth	NRCS Web Soil Survey, Ecological Site records
F06 Water Availability	Pond/stream detection (spectral water indices, object recognition)	N/A	N/A	USGS NHD (streams, lakes), High-res topo maps; State water well records (if accessible)
F07 Water Distribution	(Derived) Distance raster to nearest water point (computed from Mireye/USGS water)	N/A	N/A	N/A (derived from F06 data using GIS)

Factor	Mireye/Imagery (example fields)	USDA-ARS RAP / Veg. Databases	NRCS/Soil Data	USGS/ Hydrography/ FEMA/NOAA
F08 Precipitation	None (Mireye not used)	N/A	N/A	NOAA PRISM climate normals; NCEI 30yr normals; Climate Divisions
F09 Drought Exposure	Vegetation anomaly trends (NDVI series)	N/A	N/A	NOAA Drought Monitor; Palmer Drought Index; NCEI precip history
F10 Flood Exposure	Lidar-derived wetness indices; NDWI from imagery	N/A	SSURGO flooding frequency; hydric soil indicator	FEMA 100-yr Floodplain (FEMA), NHD (perennial vs intermittent streams)

Coverage: USDA-ARS RAP and NRCS SSURGO cover essentially all U.S. rangelands. USGS DEM and FEMA maps are national (DEM varies by state in resolution). NOAA climate/drought data covers all states.

Factor Selection Matrix

Below is a qualitative scoring of each factor on key criteria for MVP selection. This helps confirm these 10 are strong choices. (“High/Medium/Low” or Yes/No)

Factor (ID)	Scientific Importance	Shared Across Species	Differentiating Power	Remote Data Availability	Mireye Coverage	Data Reliability	MVP Priority
Terrain & Slope (F01)	High	Yes	High (cattle vs goats)	High (DEM exists)	Good (Lidar, slope)	High (terrain well-mapped)	High
Herbaceous Cover (F02)	High	Yes	Medium (all species)	High (RAP data)	High (satellite NDVI)	High	High
Shrub Cover (F03)	High	Yes	High (goat vs cattle)	High (RAP shrub data)	High (image indices)	Medium	High
Herbage Production (F04)	High	Yes	Medium	High (RAP biomass)	Medium (biomass indices)	Medium-High	High

Factor (ID)	Scientific Importance	Shared Across Species	Differentiating Power	Remote Data Availability	Mireye Coverage	Data Reliability	MVP Priority
Soil/Water Capacity (F05)	High	Yes	Low-Moderate	High (SSURGO)	Low (indirect)	High	Medium
Water Availability (F06)	Very High	Yes	Low (all need water)	Medium (GIS detect)	Medium (pond detect)	Medium (mapping gaps)	High
Water Distribution (F07)	High	Yes	Medium (cattle restricted)	Medium (GIS compute)	Low (needs water data)	Medium	Medium
Precipitation (F08)	High	Yes	Low (contextual)	High (NOAA data)	N/A	High	Medium
Drought Exposure (F09)	High	Yes	Medium (risk factor)	High (NOAA data)	N/A	High	Medium
Flood Exposure (F10)	Medium	Yes	Low	Medium (FEMA/ SSURGO)	Medium (DEM)	Medium	Low-Med

All ten factors scored high in scientific importance and availability. F05 and F10 are slightly less differentiating but still valuable context layers. All were deemed MVP priorities as they are **shared** and cover critical aspects of grazing.

Validation Guidance

To ensure the agent reflects real-world ranching, we recommend a **Validation Set** of 6–10 Texas reference ranches representing diverse conditions. These reference cases serve for sanity-checks, not for training rules. Use two modes:

- **Known-Operation Validation:** Select established ranches with known primary use (e.g. a cow-calf beef ranch, a sheep operation, a goat farm). **Blind test:** feed only the parcel data to RangeMatch (no hints of actual use). The agent should return highest suitability for the real-world use or at least not strongly contradict it. If a reference cattle ranch is rated best for goats and cattle is near zero, that flags a rule or factor error. This mode checks consistency with practical use.
- **Cross-Use Validation:** Test that the agent does not always choose cattle. For example, pick a brush-heavy ranch that in reality raises goats: the agent should prefer the Goat profile over Cow profile for this land. Conversely, a flat open range with few shrubs should rank Cow highest. This ensures discrimination between profiles.

Protocol: For each reference ranch, compile the Land Profile (factors F01–F10) from data; run all three profiles; note the ranked suitability. Compare with known operation. Any large mismatch prompts review of factor definitions or weights (or identifies cases where business decisions override soil logic). Always document: “Source Operation: cattle; Predicted Suitability: Cow=85, Sheep=45, Goat=30” (numbers just example) and annotate discrepancies.

Important: Real-world operation is not “optimal land use” ground truth. Many ranchers persist with cattle on marginal brushy land due to market or tradition. So a sheep ranch being rated moderate for cattle doesn’t prove error – just note the trade-offs. Use references mainly to catch blatant misclassifications (e.g. flooded pasture flagged as ideal for any livestock without note).

Scientific Guardrails & Versioning

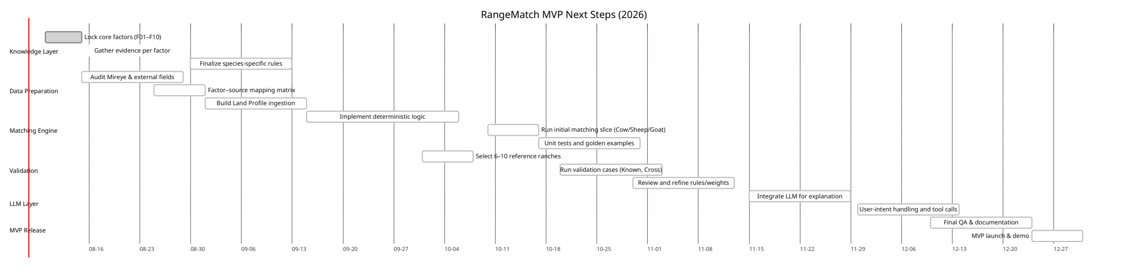
To maintain rigor and transparency, the knowledge base should follow these principles (locked by version control):

- **No Hard-Coded Universal Thresholds:** Avoid arbitrary cutoffs unless supported by evidence with scope (e.g. “slopes >30% usually reduce cattle performance” only if cited regionally). Do not apply one-size-fits-all rules nationally.
- **Data vs Inference Separation:** Land Facts (measured data) must remain distinct from Analysis Results. Inferences from LLM or rules are never written back into the Land Profile.
- **Evidence Backing:** Each rule or weight should cite a source or domain expert consensus. Mark any “expert judgment” clearly and revise as evidence emerges.
- **Uncertainty & Unknowns:** If data is missing or ambiguous (e.g. no mapped water), explicitly output `UNKNOWN` or `NEEDS_VERIFICATION`. Do not silently guess.
- **Operational vs Biological:** Keep infrastructure, finance, legal issues in a separate diligence layer. The matching engine focuses on biological/site suitability only.
- **Provenance & Versioning:** Record data source and fetch dates for every Land Fact. Version both the Operation Profiles and the rule/Evidence libraries. Output the versions in every report.

Example guardrail entries (as in config):

```
scientific_guardrails:
  - No universal numeric threshold unless regionally validated.
  - Mapped vegetation ≠ guaranteed forage (needs local knowledge).
  - Mapped water presence ≠ confirmed water supply.
  - Lack of data flags UNKNOWN; no silent assumptions.
  - Legal/infrastructure issues are OUTSIDE biological score.
```

Next Steps Timeline



YAML: Locked MVP Metadata


```yaml
shared_core_factors:
 - F01: Terrain_Slope
 - F02: Herbaceous_Cover
 - F03: Shrub_Cover
 - F04: Herbaceous_Prod_Signal
 - F05: Soil_Water_Capacity
 - F06: Water_Availability
 - F07: Water_Distribution
 - F08: Precipitation
 - F09: Drought_Variability
 - F10: Flood_Exposure

context:
 - climate_zone
 - ecoregion
 - nrcs_ecological_site

version: "MVP-2026-1.0"
locked: true
```


Sources

- USDA NRCS – Illinois Grazing Manual, Livestock Distribution (slope and water effects)【1†L39-L41】【1†L31-L37】.
- USDA NRCS – Livestock Pipeline / Prescribed Grazing (water infrastructure, grazing centers)【1†L29-L37】【8†L2-L9】.
- USDA NRCS – Range and Pasture Handbook (grazing distribution factors).
- USDA NRCS – Ecological Site Descriptions, Web Soil Survey (soil, climate, vegetation frameworks)【4†L132-L141】【10†L4-L7】.
- USDA-ARS (Agricultural Research Service) – Cattle Grazing Distribution study (cattle prefer low elevations in dry regions, avoid flooding)【4†L132-L141】.
- USDA-ARS – Multi-year Drought Study (drought impacts on forage production/quality)【6†L23-L32】.
- USDA-ARS – Rangeland Analysis Platform (RAP) (national vegetation cover and production data)【3†L198-L202】【3†L214-L223】.
- University of Minnesota Extension – Multi-species Grazing (diet preferences: cattle ~70% grass, sheep ~50%, goats ~30%)【3†L198-L202】.
- University of Idaho Extension – Multispecies Grazing (topographic and water behavior of cattle, sheep, goats; dietary differences)【3†L198-L202】【3†L214-L223】.
- Texas A&M AgriLife – Improving Rainfall Effectiveness on Rangeland (rainfall is major forage limiter in Texas)【8†L2-L9】.
- FEMA / USGS – National Flood Insurance maps; USGS NHD streams/lakes; NOAA PRISM climate normals; NOAA Drought Monitor.
